# Multi-Agent Reinforcement Learning for Air Handling Unit Control

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## 1 Working title

Multi-Agent Reinforcement Learning for Air Handling Unit Control

## 2 Background

Heating, Ventilation, and Air Conditioning (HVAC) systems account for a major share of building energy consumption and are essential for maintaining indoor thermal comfort, air quality, and ventilation performance. Among HVAC subsystems, Air Handling Units (AHUs) play a central role in regulating airflow, supply-air temperature, heat recovery, and air distribution throughout buildings.

This project focuses on a real AHU configuration derived from the KTH Live-in Lab ventilation system. The investigated AHU contains multiple interacting components, including supply and exhaust fans, rotary heat exchangers, heating coils, dampers, and various airflow, pressure, and temperature sensors. These components are thermally and operationally coupled, meaning that actuator changes may simultaneously affect airflow balance, heat recovery, energy consumption, and supply-air temperature regulation.

In practical Building Management Systems (BMS), AHUs are typically controlled using independent PID loops and rule-based supervisory logic. Although these approaches provide stable and reliable operation, they mainly optimize local control variables rather than overall system-level performance. This may lead to conservative operation, conflicting actuator behavior, and unnecessary energy consumption under changing environmental conditions.

Recent advances in reinforcement learning (RL) and deep reinforcement learning (DRL) have shown promising potential for intelligent HVAC control. However, most existing studies focus on zone-level temperature regulation or whole-building optimization using simplified simulation environments. Relatively limited attention has been given to actuator-level coordination inside real AHU systems using operational measurements and physical constraints.

This thesis therefore investigates a Multi-Agent Reinforcement Learning (MARL) framework for actuator-level AHU coordination. Individual learning agents will be assigned to major AHU actuators, such as supply fans, exhaust fans, heating valves, and rotary heat exchangers, while coordination between agents will aim to improve energy efficiency, airflow regulation, and operational stability.

Operational data from the KTH Live-in Lab system, including temperature, airflow, pressure, and actuator measurements, will be used for system analysis and controller evaluation.

# 3 Research question

The main research question of this thesis is:

> How can a multi-agent reinforcement learning framework coordinate multiple coupled AHU actuators in a real ventilation system to improve energy efficiency while maintaining supply-air temperature, airflow balance, ventilation requirements, and operational constraints?

The thesis also investigates several sub-questions:

- How should the AHU system be decomposed into interacting control agents?
- Which operational measurements and actuator signals are most relevant for defining the MARL state representation?
- How should reward functions balance energy consumption, thermal comfort, airflow regulation, and control stability?
- Can coordinated actuator-level MARL outperform conventional rule-based or PID-based control strategies under realistic operating conditions?

# 4 Hypothesis

Compared with conventional rule-based and independent PID-based AHU control strategies, a coordinated multi-agent reinforcement learning framework is expected to:

- reduce unnecessary heating and fan energy consumption;
- improve coordination between thermally and operationally coupled actuators;
- maintain more stable supply-air temperature regulation;
- reduce conflicting actuator behavior during changing environmental conditions;
- improve overall AHU operational efficiency while satisfying ventilation and operational constraints.

It is further expected that actuator-level coordination through decentralized learning agents can provide better scalability and flexibility than fully centralized control approaches for complex AHU systems.

# 5 Research method

The project will combine AHU system analysis, operational data analysis, reinforcement learning, and controller evaluation.

First, the AHU topology and operational measurements from the KTH Live-in Lab system will be analyzed to identify major actuators, measurable states, operational constraints, and coupling relationships between subsystems. Available measurements include temperature signals, airflow measurements, pressure measurements, and actuator control signals.

Second, the AHU control problem will be formulated as a Markov Decision Process (MDP). The system state is expected to include temperature, airflow, pressure, and actuator-related measurements, while the action space will consist of controllable AHU actuator signals such as fan speed, heating valve position, and rotary heat exchanger operation.

Third, a Multi-Agent Reinforcement Learning (MARL) framework will be investigated, where individual agents are associated with major AHU actuators. Different reward formulations and coordination strategies may be explored to balance energy efficiency, airflow regulation, supply-air temperature stability, and operational constraints.

Depending on data availability and project scope, the controller may be evaluated using simulation environments, historical operational data, or simplified data-driven models. The proposed approach will be compared with conventional rule-based or PID-based control strategies using metrics related to energy consumption, temperature regulation, airflow stability, and overall operational performance.

The final outcome of the project will include system analysis, MARL framework development, controller evaluation, and documentation in the final master's thesis.

## 6 Background of the student

Haojie has an academic background in communication systems, networking, distributed systems, and machine learning. Previous coursework and project experience include reinforcement learning, data-driven system analysis, IoT systems, distributed architectures, and Python-based implementation of intelligent systems.

Haojie also has experience working with real operational data, embedded systems, and system-level experimentation. In addition, previous work involving machine learning pipelines, control-related problems, and data analysis provides a suitable foundation for reinforcement learning-based HVAC control research.

The project also aligns with Haojie's interest in intelligent control systems, multi-agent coordination, and practical deployment of learning-based methods in real engineering systems.

## 7 Suggested examiner at KTH

Still Finding.

## 8 Suggested supervisor at KTH

Youssef Elomari

## 9 Resources

The project builds upon operational measurements and AHU system configurations provided by the KTH Live-in Lab building ventilation system.

Available resources include:

- AHU operational data;
- actuator control signals;
- airflow, pressure, and temperature measurements;
- system topology and sensor layouts;
- existing building management infrastructure;

- supervision and domain expertise related to HVAC systems and building energy control.

Software tools such as Python-based reinforcement learning frameworks, data analysis libraries, and simulation environments will also be used during the project.

# 10 Eligibility

Haojie has completed 67.5 ECTS credits within the Master's Programme in Communication Systems at KTH, including courses related to machine learning, advanced internetworking, IoT systems, communication system design, and research methodology.

These completed courses provide the necessary background for the proposed degree project in reinforcement learning-based HVAC control.

The remaining registered courses are planned to be completed during the current academic period before the beginning of the thesis project. Therefore, Haojie is expected to fulfill the eligibility requirements for starting the degree project.

# 11 Study Planning

Haojie is currently registered in the following remaining courses:

- IK2221 – Networked Systems for Machine Learning;
- IK2227 – Network Systems with Edge or Cloud Datacenters;
- ME2062 – Technology-based Entrepreneurship.

These courses are expected to be completed during the current academic period, before June 2026.

The degree project is planned to be one of the final components of the Master's Programme in Communication Systems.